# Lab 7: Fuzzing for Input Validation Bugs

Name: Maansi S | SRN: PES1UG24AM913 | Module: Income Tax Calculator (income_tax_parser.py)

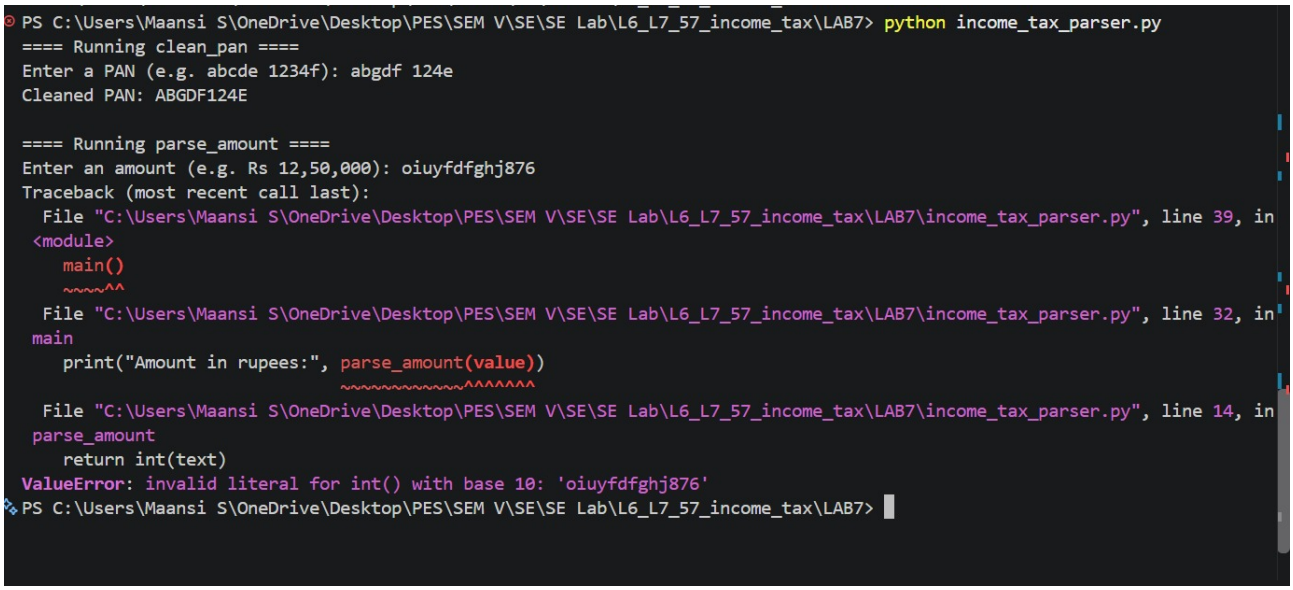
## Summary

The module **income_tax_parser.py** contains three functions, **clean_pan**, **parse_amount** and **parse_tax_slab**, that assume every input is well formed. Property-based fuzzing with Hypothesis (`st.text() | st.none()`) showed that all three crash on unexpected input. The module was then hardened with type checks and validation, and the fuzz tests were re-run and pass.

## Bugs found and fixes

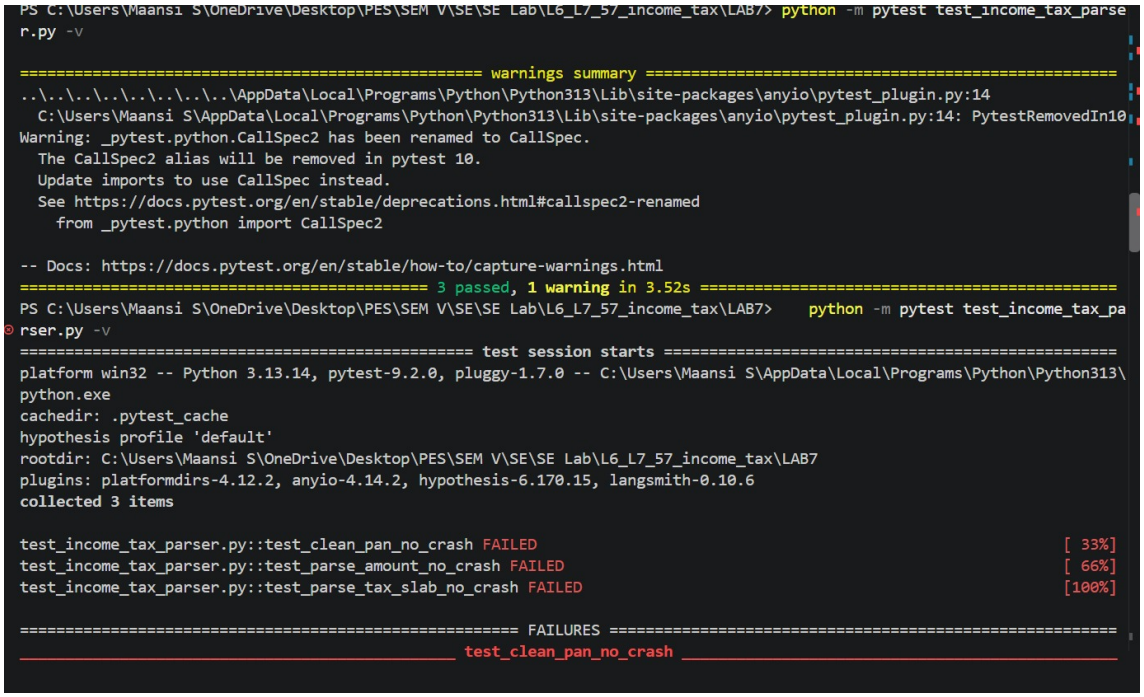
| Function | Failing input | Exception | Fix |
|---|---|---|---|
| clean_pan | None | AttributeError | Return None if not a string; strip all whitespace |
| parse_amount | None | AttributeError | isinstance check |
| parse_amount | "", "abc", "Rs" | ValueError | Validate digits with regex before int(); return None |
| parse_tax_slab | None | AttributeError | isinstance check |
| parse_tax_slab | "", "1-2", "1:2:3" | ValueError | Check exactly one ":" and one "-" before unpacking |
| parse_tax_slab | "a-b:c" | ValueError | Validate each part as digits; check low <= high, rate <= 100 |

## SS1: python income_tax_parser.py on the buggy module

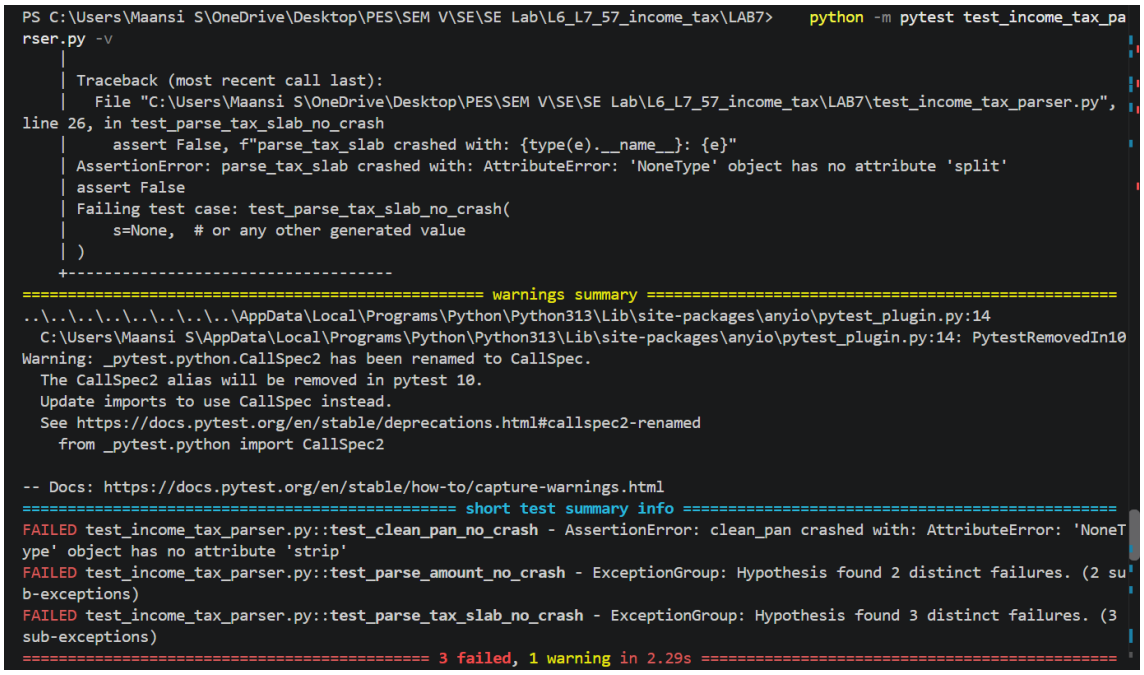



Entering the non-numeric amount "oiuyfdfghj876" crashes parse_amount with an unhandled ValueError (int() on invalid text).

## SS2: pytest test_income_tax_parser.py on the buggy module

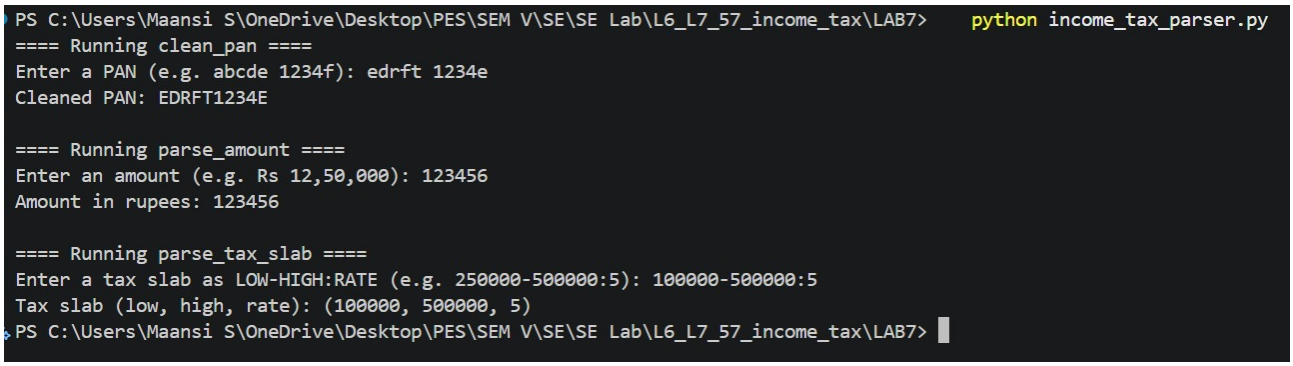



Test run header: all three fuzz tests FAILED on the buggy module.



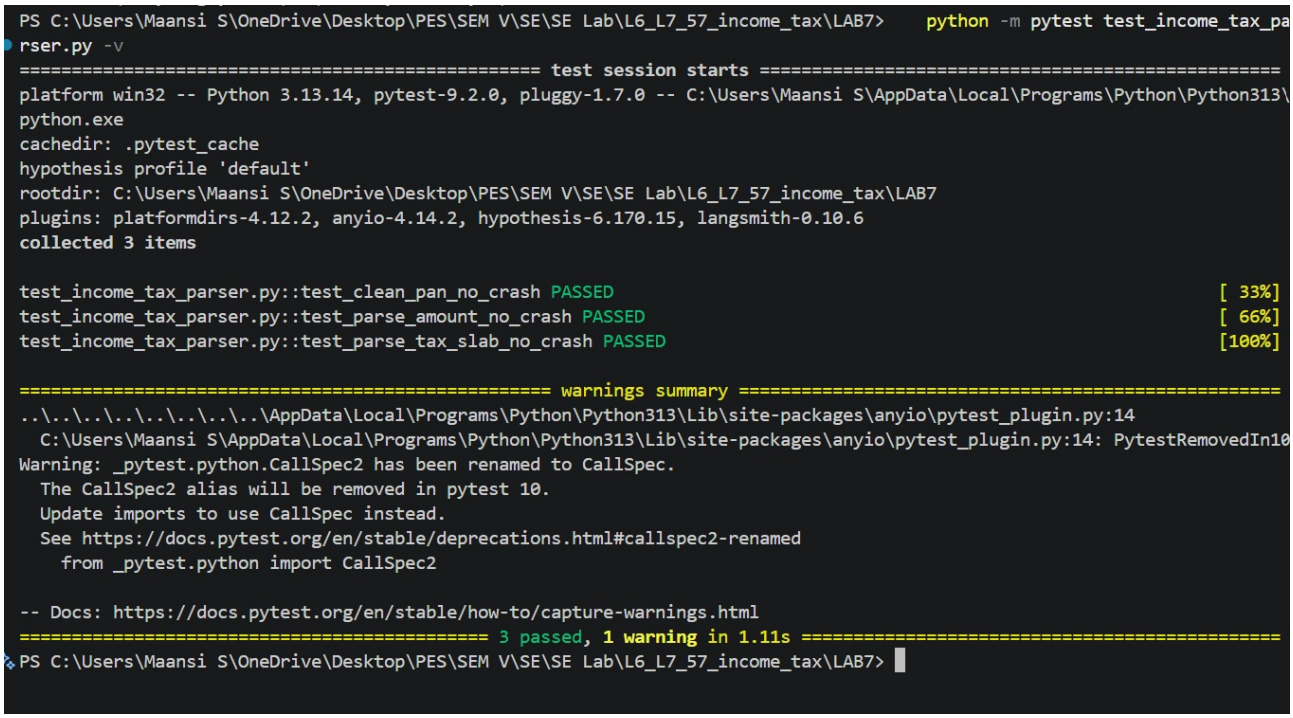

Failure details: Hypothesis found the falsifying example s=None, giving AttributeError ("NoneType" has no attribute "strip"/"split"). Summary: **3 failed** (clean_pan, parse_amount with 2 distinct failures, parse_tax_slab with 3 distinct failures).

## SS3: python income_tax_parser.py on the fixed module




The fixed module runs cleanly on valid inputs: PAN cleaned, amount parsed, slab parsed.

## SS4: pytest test_income_tax_parser.py on the fixed module




All three fuzz tests PASS (3 passed) against the hardened module.